

OS [Lec-4]

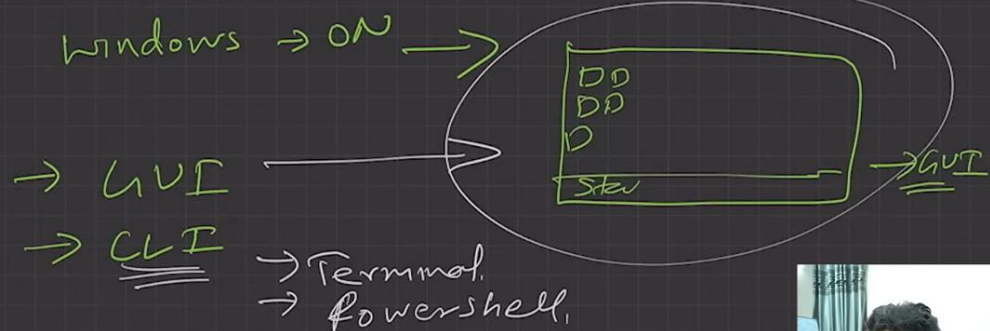
Components OS :-

- ① User space ✓ → APP are run here
- ② kernel ✓ → access to underlying hardware.



① User space

- No hardware access
- Convenient environment user apps.

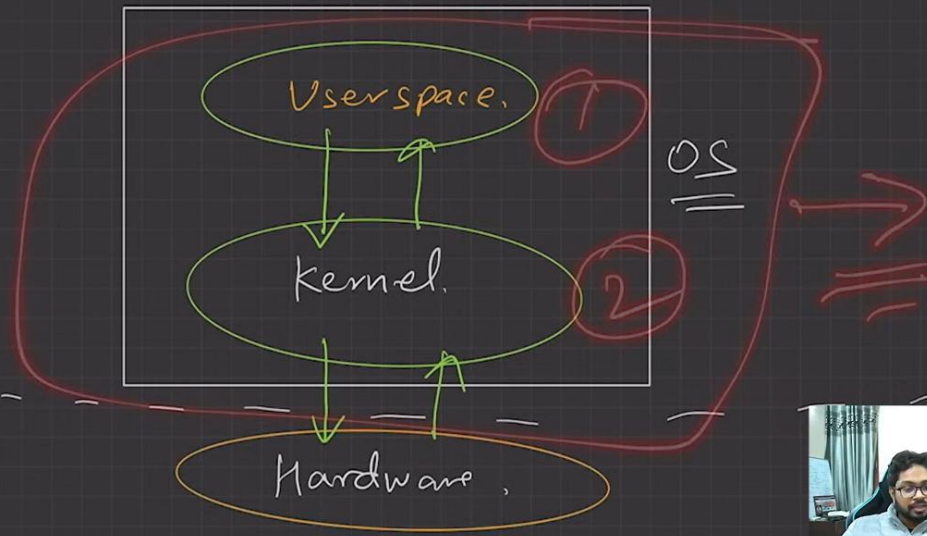


→ Interacts with kernel

② kernel :- Heart of OS

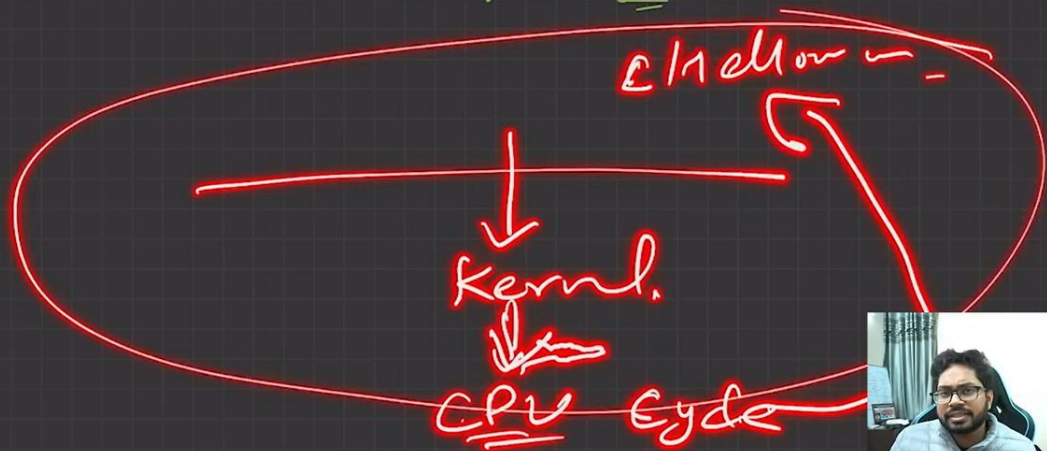
- Interacts with hardware

- Interacts with Hardware



eg. helloworld.sh | -

/helloworld.sh



Kernel :- $f^n(s)$

① Process mgmt :-

- Process creation, termination
- Process & thread schedule.
- Process sync.

→ Process communication :-

$P_1, P_2 \rightarrow$ Isolated
memory



② Memory mgmt :-

- Allocate / deallocate
- Free space mgmt.

③ File mgmt :-

- create / delete file
- directory mgmt

④ I/O Mgmt :-

→ mgmt & controlling of all I/O devices.

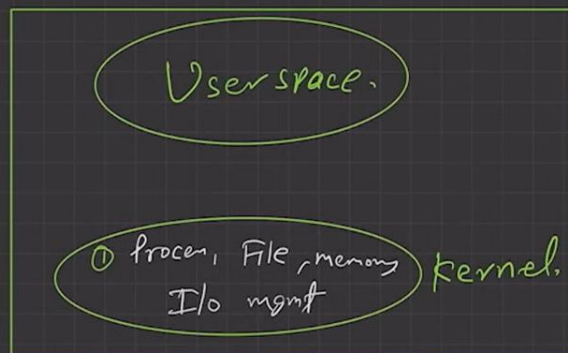
4.1) Spooling

4.2) Buffering

4.3) Caching



① Monolithic Kernel:-



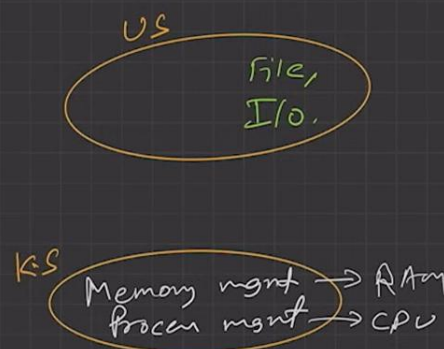
→ User mode
Kernel mode

OS



Adv → → Fast communication b/w diff components of kernel.
 disadv → Kernel Bulky
 → Less reliable.
 eg.- Linux & Unix, MS/DOS

② Micro Kernel:-



Adv. → Less Bulky
 → More Reliable.
 → Stable

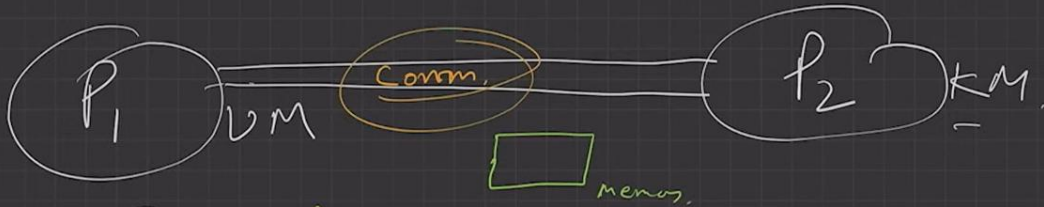
Disadv.:-
 → Performance less
 → Overhead → UM/KM

US → K.S → U.S → K.S → U



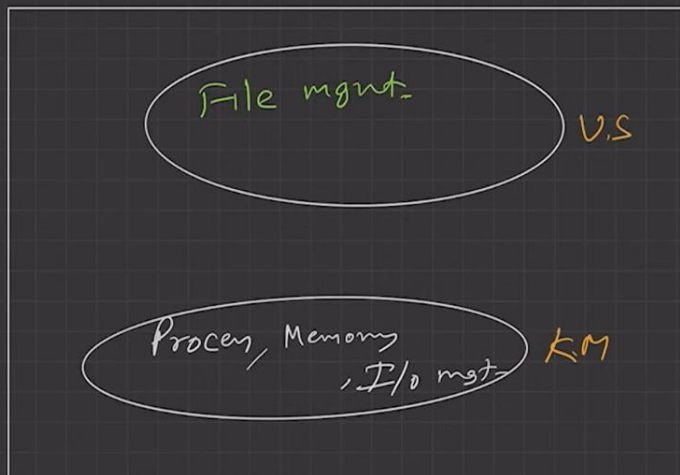
Interprocess Communication

- ① $VM \neq KM \rightarrow \underline{Comm.} ??$
 $\rightarrow \underline{IPC} \rightarrow \text{Interprocess} - \underline{Communicator}$



- ① Shared memory -
② Message passing -

- ③ Hybrid kernel.
— Combined approach.



eg. Mac OS
Windows NT
7+